MySQL Fundamentals

This guide introduces relational databases and MySQL for beginners using Windows and Visual Studio Code.

1. What is a Database?

A database stores information in an organized way so it can be searched, updated and managed efficiently. Unlike JSON files, databases are designed to handle large amounts of data, multiple users and complex searches.

2. Why MySQL?

MySQL is a Relational Database Management System (RDBMS). Data is stored in tables made of rows and columns. Tables can be related using keys.

3. Important Terms

Table: stores similar data.
Row (Record): one item.
Column (Field): one property.
Primary Key: uniquely identifies each row.
Foreign Key: links one table to another.

4. Installing MySQL (XAMPP)

1. Download XAMPP.
2. Install with MySQL selected.
3. Open XAMPP Control Panel.
4. Start MySQL.
5. Open http://localhost/phpmyadmin in your browser.

5. Using VS Code

Install extensions:
• SQLTools
• SQLTools MySQL/MariaDB Driver
Create a new connection using host=localhost, port=3306, username=root and your password (or blank if using XAMPP defaults). Test the connection.

6. Create a Database

```
CREATE DATABASE school;
USE school;
```

7. Create a Table

```
CREATE TABLE students(
id INT AUTO_INCREMENT PRIMARY KEY,
name VARCHAR(100),
age INT
);
```

8. Insert Data

```
INSERT INTO students(name,age)
VALUES('John',15),('Mary',16);
```

9. Read Data

```
SELECT * FROM students;
SELECT name,age FROM students;
```

10. Update Data

```
UPDATE students
SET age=17
WHERE id=2;
```

11. Delete Data

```
DELETE FROM students
WHERE id=1;
```

12. Filtering

```
SELECT * FROM students WHERE age>=16;
SELECT * FROM students ORDER BY age DESC;
SELECT * FROM students LIMIT 5;
```

13. Aggregate Functions

COUNT(*), AVG(age), MAX(age), MIN(age), SUM(age) help summarize data.

14. GROUP BY

GROUP BY groups rows before calculating values.
Example:
SELECT age, COUNT(*) FROM students GROUP BY age;

15. JOIN

Use JOIN to combine related tables.
Example: students, courses and enrollments can be joined to show which course each student takes.

16. Mini Project

Create a School Database containing students, teachers and courses. Insert at least 10 records into each table. Write queries using SELECT, WHERE, ORDER BY, GROUP BY and JOIN.